

EngineSoundID: Real-Time Car Engine Type Identification from Sound

Student Information

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Project Overview

For this project, a new app will be proposed; EngineSoundID, a real-time audio recognition system that identifies a car's engine type from its sound. The problem it addresses is that, while expert listeners can often distinguish engines by ear, there is currently no widely available app that can automatically recognize engine characteristics from audio. This capability is important for car enthusiasts, automotive diagnostics, and intelligent transportation systems. For example, engine sounds carry information about a vehicle's performance and condition [2], which could be leveraged for monitoring engine health or simply for identifying an unknown car by its sound.

The solution is a smartphone or PC-based application that listens to an engine's audio (either live via microphone or from a recording) and instantly classifies the engine type. Engine type here refers to categories such as engine configuration or vehicle class (e.g. V8 gasoline, V6, inline-4, diesel, electric motor, etc.). The app will analyse short sound clips and output the predicted engine category in real-time (within a few seconds of audio input). The system will combine classical signal processing and modern machine learning to achieve reliable recognition. If successful, EngineSoundID would essentially function similar to Shazam, being able to identify what's under the hood just from the sound.

Background and Motivation

As a fan of cars, and especially loud engines, I'm very interested to work on a project that could have real use for myself and other people. Often at car meets or on the street, I find myself guessing a car's engine type by sound alone, which can be difficult especially on modified cars. On a practical level, such technology could assist mechanics and smart vehicles: research shows engine sounds can reveal performance issues or malfunctions [2]. An app that recognises engine types could eventually be extended to detect engine problems or enhancements from the audio, blending my passion for cars with engineering innovation.

From a technical perspective, identifying engines by sound is a challenging task. Engine noises are complex, containing a mix of combustion tones, exhaust harmonics, and mechanical vibrations across a wide frequency range [3], including background noise (wind, road, other vehicles) that often interferes, making feature extraction and recognition difficult [2]. Despite these challenges, prior studies indicate it's feasible: researchers as far back as the 1970s attempted to classify vehicle sounds using cepstral analysis, and modern efforts use advanced techniques to improve accuracy [2]. For example, Mel-Frequency Cepstral Coefficients

(MFCCs) have been applied to engine sound analysis. Kemalkar and Bairagi [1] used MFCC-based features to successfully detect engine faults from audio signals, showing that engine sound signatures can indeed be captured and analysed [1]. More recently, deep learning approaches have emerged: a 2022 study [2] achieved about 84% accuracy in recognizing different car engine sound conditions using a specialized convolutional neural network [2]. This demonstrates both the potential of machine learning and the need for substantial data and robust algorithms to reach high performance.

Proposed Methodology

Development Platform: The project will be implemented primarily in Python, taking advantage of its rich libraries for audio processing and machine learning. Libraries such as *Librosa* (for feature extraction), *pydub/FFmpeg* (for audio handling), and frameworks like *TensorFlow* or *PyTorch* (for neural networks) will be used.

Data Collection and Preprocessing: The first step is to build a labeled dataset of engine sounds. We plan to gather audio samples of at least 10 distinct engine types (covering various cylinder configurations and vehicle categories). These will be obtained by scraping public videos (e.g. YouTube clips of engine sound tests or exhaust notes) and recording personal audio from vehicles where possible. Each clip will be labeled with the engine type (for instance, “V8 petrol,” “V6,” “inline-4,” “diesel truck,” “electric motor,” etc.). I will ensure clips are trimmed to a consistent length (a few seconds of steady engine sound) and converted to a standard format (e.g. 44.1 kHz mono WAV). Some preprocessing will be applied to improve quality: noise reduction or band-pass filtering to focus on engine-relevant frequencies. Data augmentation may also be used (e.g. adding slight noise or pitch shifts) to increase diversity if the dataset is small [3].

Prototype 1 – Classical Signal Processing: The initial prototype will use a more standard approach for pattern recognition on the audio signals. First, I will extract key acoustic features from each engine audio sample. In particular, MFCCs will be computed, as they effectively capture the timbral and frequency content of sounds and are widely used in audio recognition [2]. Using these features, a comparison algorithm will identify the closest matching engine type. One method to be explored is Dynamic Time Warping (DTW), a technique that aligns sequences of feature vectors in time. DTW can measure similarity between an unknown engine sound and reference templates even if the sounds have different durations or slight speed variations. This approach was historically used in speech recognition and is suitable for matching engine sound patterns.

Alternatively, I will investigate an audio fingerprinting strategy inspired by the Shazam music recognition algorithm. Audio fingerprinting involves detecting robust time-frequency landmarks in the signal and using a hash lookup to identify matches [4]. Implementing a simplified fingerprinting system could allow real-time identification by matching an incoming sound’s “fingerprint” against a database of known engine fingerprints. Prototype 1 will thus likely consist of a feature extraction module (MFCCs and possibly a fingerprint generator) and a matching/classification module (using DTW or nearest-neighbor search in feature space). This prototype serves as a baseline; it is relatively interpretable and may work well if engine sounds have clearly distinguishable feature patterns.

Prototype 2 – Machine Learning (CNN-Based): If the previous approach struggles to achieve the desired accuracy, we will develop a second prototype using machine learning. In Prototype

2, a CNN will be trained to classify engine sounds. One effective technique is to convert audio into a spectrogram or MFCC heatmap image and use a CNN to recognize patterns in these time-frequency representations, and have been used before [3]. The CNN architecture might take as input a mel-spectrogram image and output one of the engine type classes, and will need to be relatively shallow CNN (given the limited data), or use a pre-trained audio model (like a VGG network) with transfer learning to leverage existing learned features. Data augmentation (random noise, time-shifting, etc.) will be applied during training to expand the effective dataset and reduce overfitting. If the collected dataset is too small (only a few samples per class), I may rely primarily on Prototype 1 and treat Prototype 2 as an experimental extension.

Throughout development, there will be testing on the system's ability to identify engine sounds under various conditions. Python's flexibility will allow quick prototyping and integration of these components. This will also consider computational efficiency, aiming to keep the processing real-time. This might involve using short audio buffers (e.g. 1-2 seconds of sound) and ensuring algorithms (DTW or CNN inference) are optimized for speed, potentially by using lower-dimensional feature representations or hardware acceleration in the case of CNNs.

Expected Outcomes/Objectives

- **A successful product** - The system should at least distinguish a basic set of engine categories with reasonable consistency, including an unknown sound.
- **Good performance** - aim to reliably classify at least 10 different engine sound classes with a high accuracy. A quantitative goal is to achieve around 80% accuracy or above in identifying the engine type under test conditions, similar to other research [1] [3]. Comparison between prototype 1 and 2 can also be down if both are completed.
- **App** – make an actual app if possible with time using the software
- **Test** – work on unknown engine sound and see if it can determine close to engine type
- **Environment** – work even in poor, loud sounding environments.

Timeline

- **Week 6:** Literature review on engine sound recognition; collect dataset from YouTube and personal recordings
- **Week 7:** Implement Prototype 1 – feature extraction (MFCCs, spectral features) and similarity matching (DTW/fingerprinting).
- **Week 8:** Test and refine Prototype 1 on dataset, tune parameters; collect more samples if needed.
- **Week 9:** Begin Prototype 2 – prepare data, generate spectrograms, design small CNN model.
- **Week 10:** Train and evaluate CNN; compare Prototype 2 accuracy with Prototype 1.
- **Week 11:** Integrate chosen approach into a simple app interface; enable real-time audio input/output.
- **Week 12:** Robustness testing under noise; optimize code for speed; finalize results and documentation.
- **Week 13:** Deliver final demo; submit project report and GitHub documentation.

References

1. A. K. Kemalkar and V. K. Bairagi, "Engine fault diagnosis using sound analysis," in *Proc. 2016 Int. Conf. on Automatic Control and Dynamic Optimization Techniques (ICACDOT)*, Pune, India, 2016, pp. 943–946.
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4. A. L. Wang, "An Industrial-Strength Audio Search Algorithm," in *Proc. International Symposium on Music Information Retrieval (ISMIR)*, Washington, D.C., 2003.